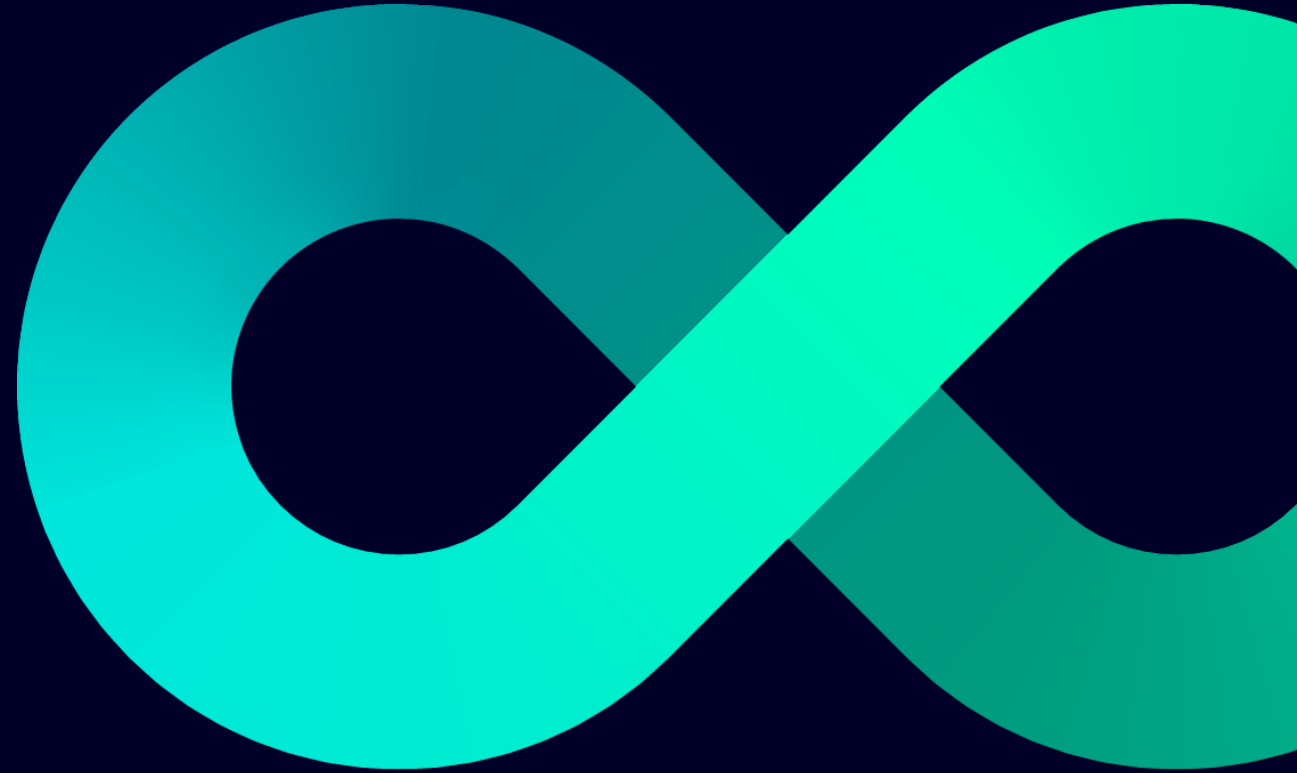


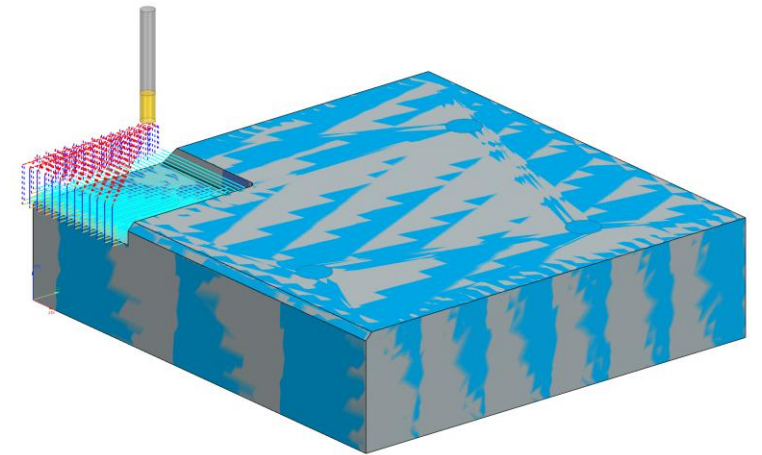
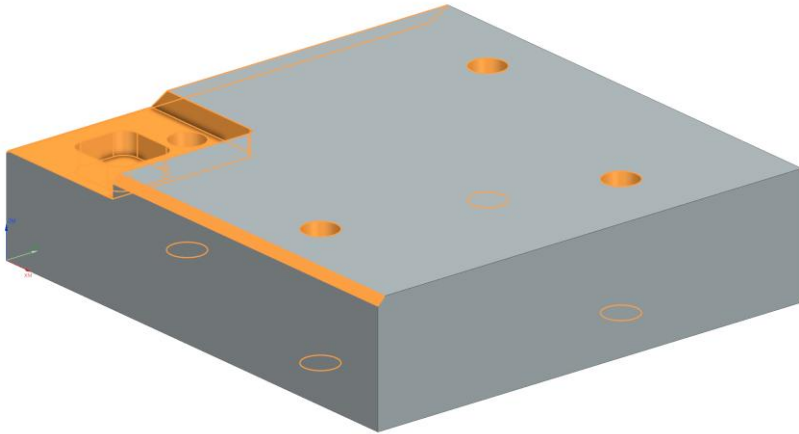
ASME 2026 Student Hackathon Problem 1

Aug 13, 2026



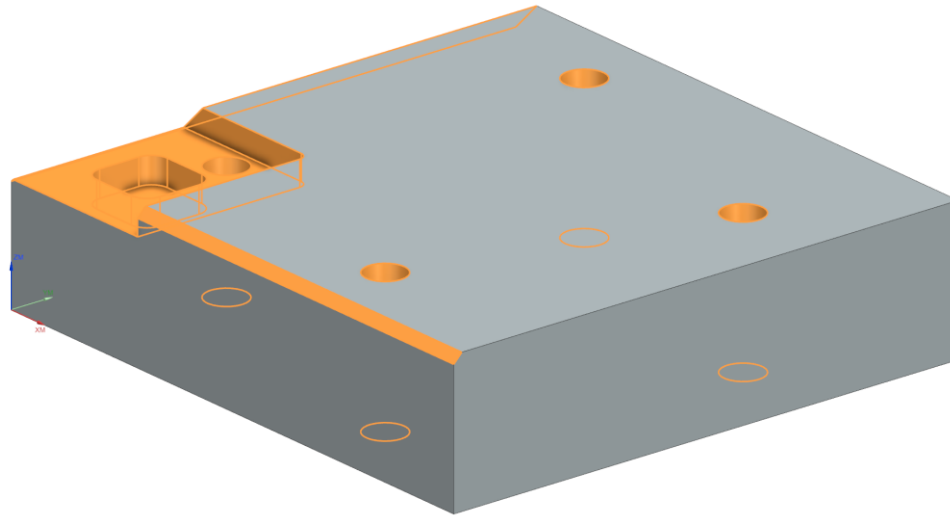
Problem Statement

- Given a part, predict the sequence of operations needed to machine it from a prismatic block using 2.5 axis CNC machining. Predict the IPW after each operation. Define the tool type, tool size, and tool path needed for each operation.



Why is it Challenging?

- Dynamic problem, not static
- Every step depends on the previous step(s)

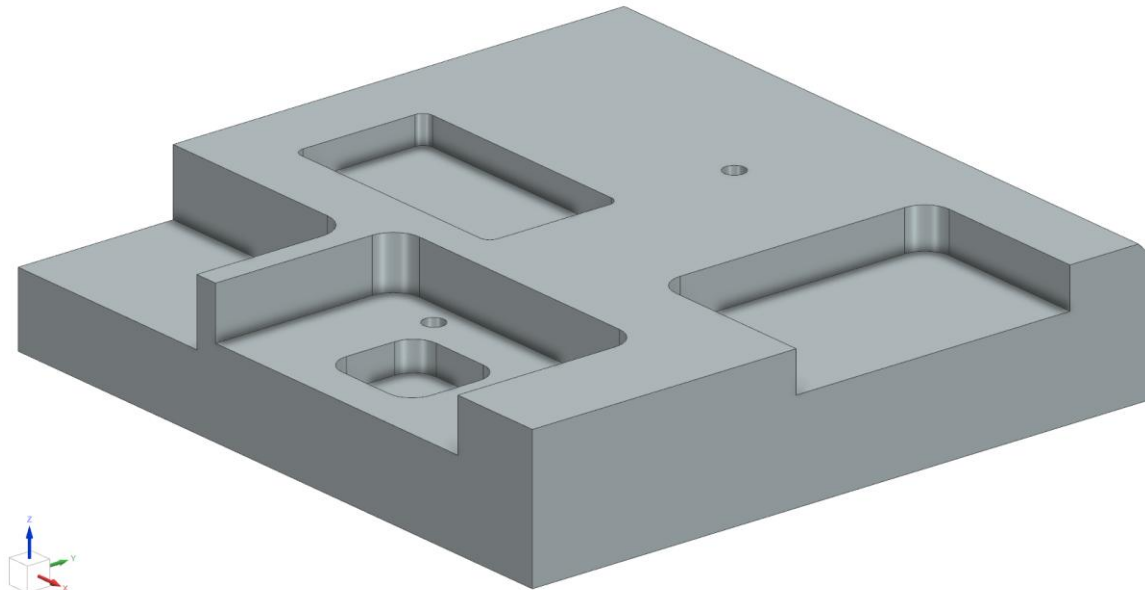
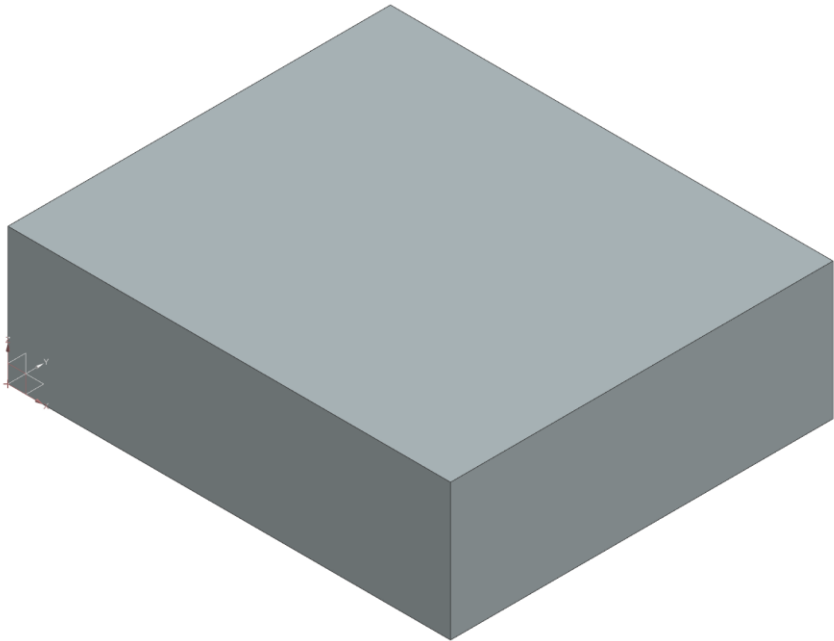


Dataset

- MachinePlan-10k
- Available at: [10.5281/zenodo.21653081](https://doi.org/10.5281/zenodo.21653081)
- A separate tutorial on dataset available
- Contains part designs as Brep (.stp), IPWs as .stl
- .txt file for each operation detailing operation type and subtype, tool type, tool size
- Wireframe CAD Images under third angle projection (.png)
- .ptp file with tool path as NC Code

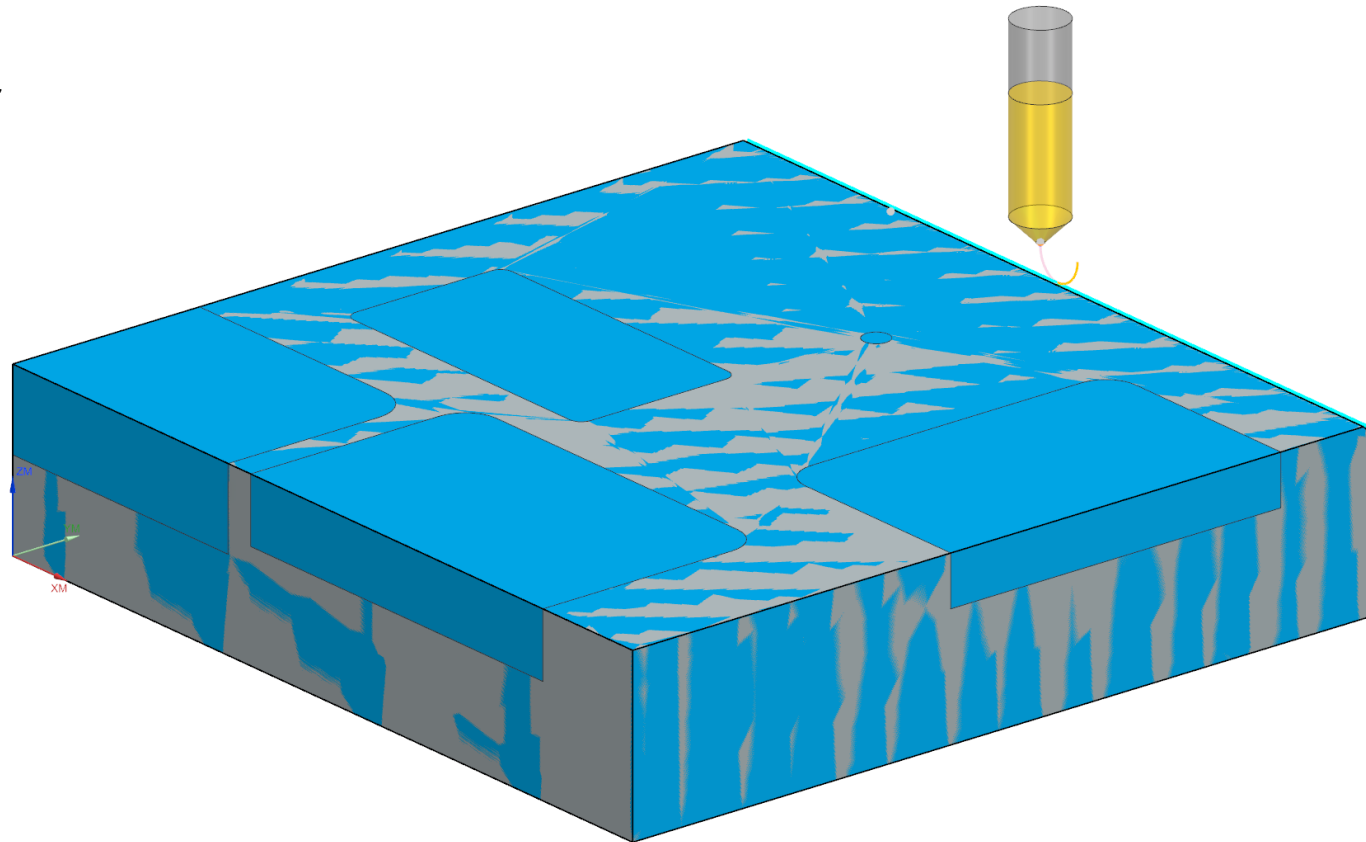
Sample Sequence

Given:
BRep (.stp)



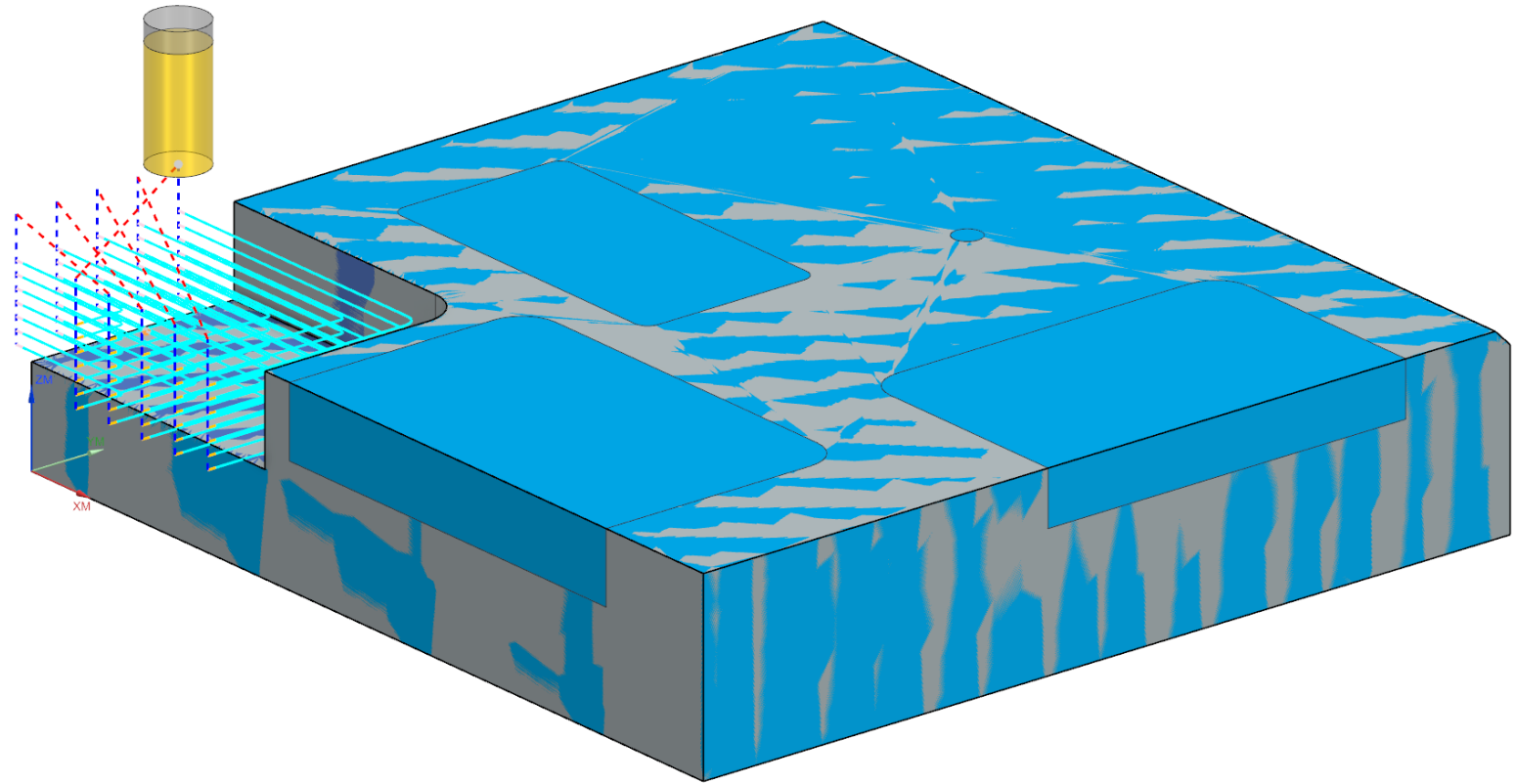
Sample Sequence: Operation 1

Main label: mill_contour
Sub label: AREA_MILL
Tool type: chamfer_mill
Tool diameter: 20 mm



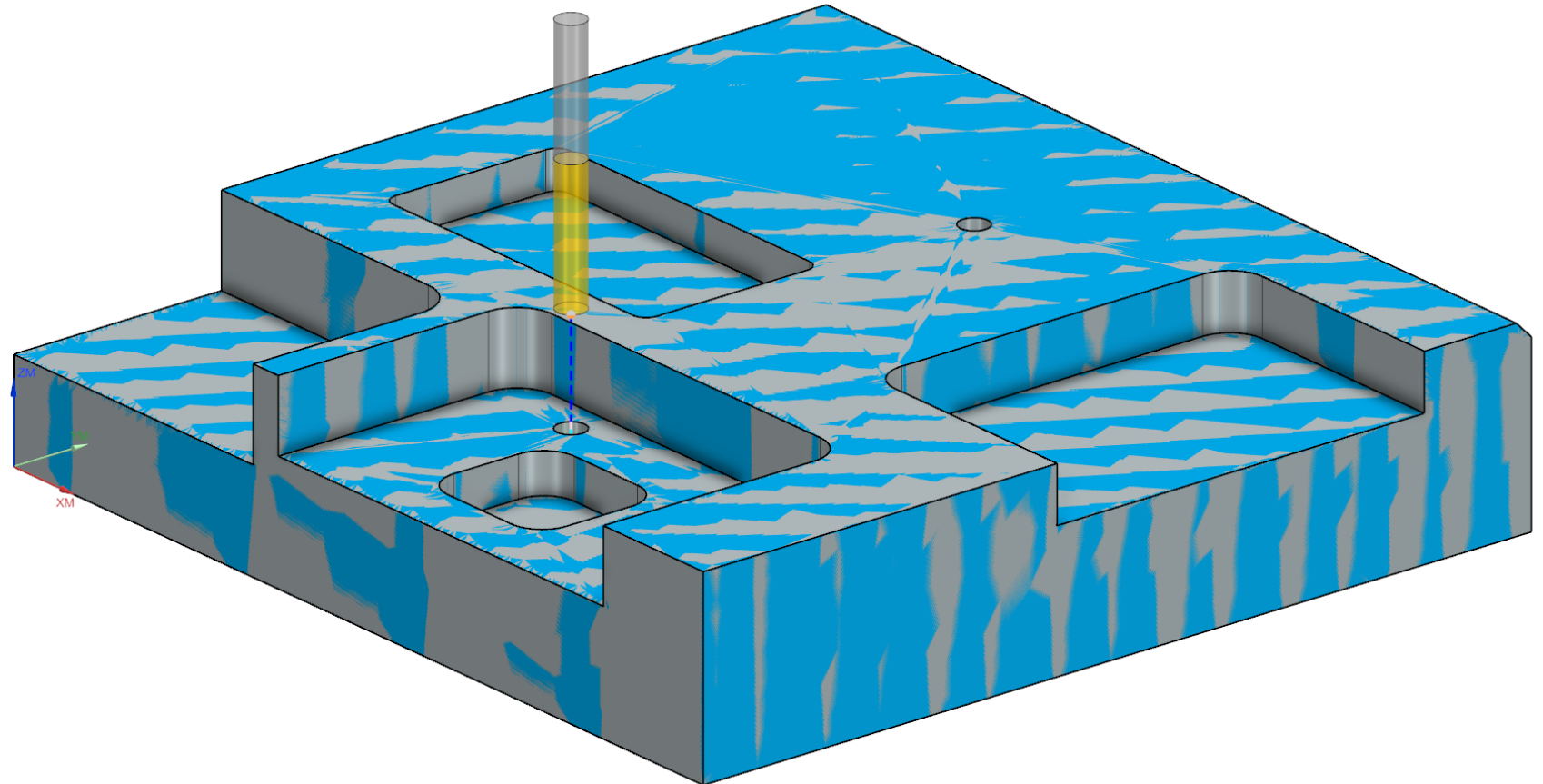
Sample Sequence: Operation 2

Main label: mill_planar
Sub label:
FLOOR_WALL
Tool type: end_mill
Tool diameter: 20 mm



Sample Sequence: Operation 10

Main label: hole_making
Sub label: DRILLING
Tool type: twist_drill
Tool diameter: 9.8 mm



Tasks

- Easy: Predict the sequence of operations
 - Each operation is recognized by two labels, main label and sub label
 - Sequence and length are both important
- Medium: Predict the IPW after each operation
 - Correct sequence is necessary to predict the IPW
 - Features can be nested or interacting; What needs to be removed depends on what has been removed
- Hard: Predict the tool type, tool size, and tool path

Deliverables

- Follow the templates, verify using helper script
- Easy:
 - Sequence of operations (.json)
- Medium:
 - One IPW mesh per operation (.stl)
- Hard
 - One tool type and tool size per operation (.json)
 - Tool path and machining parameters as .ptp file

Evaluation

- Easy: Normalized Levenshtein distance, F1 score
- Medium: IOU
- Hard : Volumetric IOU cut by tool paths, overcut and undercut %
- Detailed information available in the rubric document on Github
- Second stage evaluation will also value the methods, approaches, scientific rigor, computational efficiency etc.

Limitations and Disclaimer

- Machining can be done in multiple ways
- Same operation can be done in different number of steps
- Goal is to learn the *optimal sequence* generated by deterministic rules within Siemens Designcenter NX

Questions?

- Dataset: <https://doi.org/10.5281/zenodo.21653081>
- Github: [athulcd/ASME-CIE-Student-Hackathon-2026-Problem-1](https://github.com/athulcd/ASME-CIE-Student-Hackathon-2026-Problem-1)
 - Tutorial
 - Helper scripts
 - Dataset description